Lab 10

PH142 Fall 2026

Announcements

- **Lab10:** due Sat 11/14 at 11:59 AM (noon)
- **No quiz this week!**

Week 13 Lecture Review

Inference for Regression

Assumptions:

1. **Linearity:** The relationship between x and y is linear in the population
2. **Normality of Residuals:** Variable y varies Normally around the line of best fit
3. **Independence:** Observations are independent
4. **Constant Variance:** The standard deviation of the responses is the same for all values of x

Week 13 Lecture Review

Evaluating Assumptions

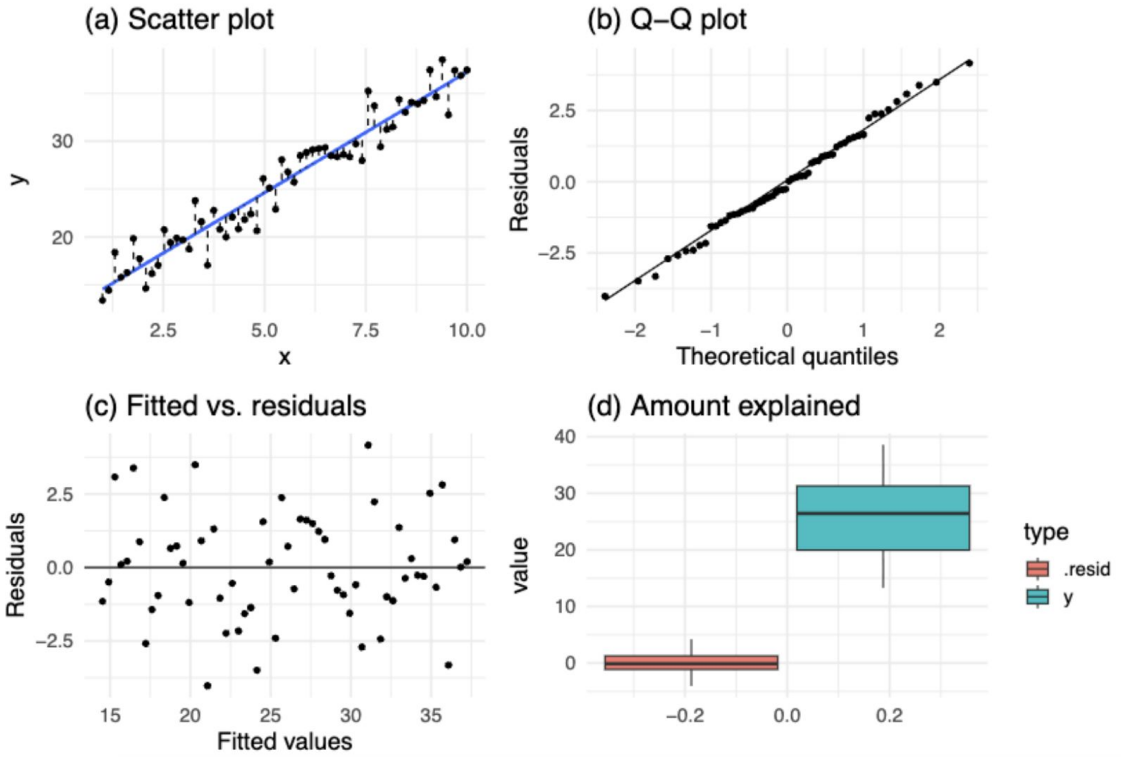
We can use different diagnostic plots to test our assumptions:

	What to Look For
Scatter plot of data w/ regression line	No trend, residuals are sometimes positive and sometimes negative
Q-Q plot of residuals	Residuals closely follow the line
Scatter plot of fitted vs. residuals	No trend, random scatter
Box plot of y vs. residuals	IQR of residuals box smaller than y box

Week 13 Lecture Review

Good Example

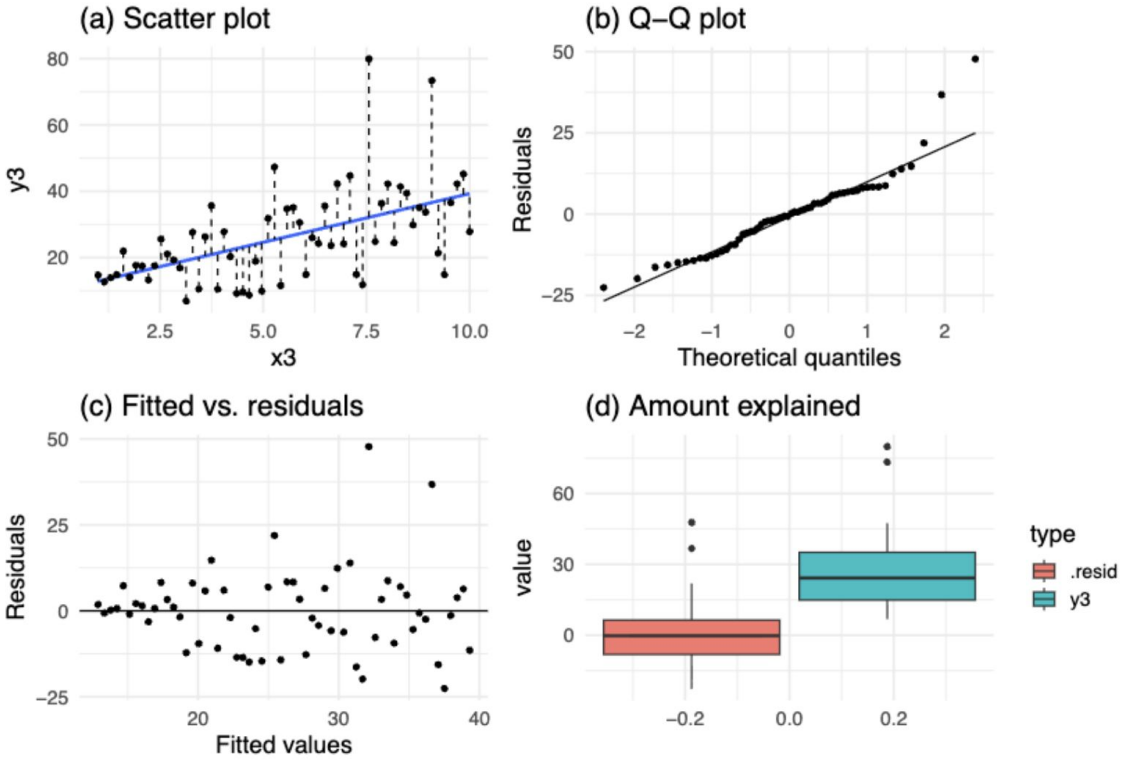
(Assumptions met)



Week 13 Lecture Review

Bad Example

(Assumptions not met)



LAB 10 Walkthrough

Lab Submission

- Follow the directions on the LAB10 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
  - Copy and paste the given line into the terminal
  - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**